

Identification of QTLs for salt tolerance at germination and seedling stage of *Sorghum bicolor* L. Moench

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Received: 13 August 2013 / Accepted: 24 October 2013 / Published online: 1 November 2013
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Abstract Salt stress is a major limitation for crop production in saline soil. For investigating genetic mechanism of salt tolerance at germination and seedling stage, and improving salt tolerance of sorghum, 181 recombinant inbred lines derived from Shihong137 and L-Tian were used in this study. Quantitative trait loci (QTLs) for three traits at germination stage and nine traits at seedling stage were analyzed. 12 and 29 QTLs were identified at germination and seedling stage, respectively. Only *qGP7-1* for germination percentage was found consistently under control and 2.0 % NaCl stress. Other QTLs detected under salt stress were salt specific expression. Six major QTLs were identified, and positive effects were all from Shihong137 except that of *qRL10-2*. Five chromosome regions controlling more than one trait simultaneously were found under salt stress. The results demonstrated that salt tolerance at germination and seedling stage of sorghum was a complex quantitative trait and controlled by multiple genes. However, six major QTLs and five chromosome regions played crucial role in salt tolerance of sorghum, which could be applied in marker assisted selection and in further investigation for salt tolerance.

Keywords Sorghum · Salt tolerance · QTLs · Germination · Seeding

Introduction

Saline soils are distributed throughout the world (Brady and Weil 2002), and a total land area of 831 million hectares is salt affected (KinfeMichael and Melkamu 2008). Soil salinity is one of main obstacles limiting crop productivity worldwide (Zhu 2001). Sorghum (*Sorghum bicolor*) is the fifth most important cereal crop of the world after maize, wheat, rice, and barley (www.fao.org), and is a major source of food, feed and fodder in semi-arid tropics (Léder 2004). It is tolerant to numerous biotic and abiotic stresses and moderately salt tolerant (Francois et al. 1984). Therefore, genes conferring resistance to biotic as well as abiotic stresses may be located in sorghum genome. Furthermore, governments and researchers paid more attention to develop renewable energy resources with increasing pressure from energy supply. Sweet sorghum could yield relatively high biomass growing in saline and alkaline land as well as barren land, where the major food crops could not grow well (Rooney et al. 2007; Wortmann and Regassa 2011). As a good candidate crop for biofuel production without arable land competition to major food crops, interest has been focused on sorghum.

The effect of salinity on plant growth is a complex syndrome, which causes osmotic stress, ion toxicity and mineral deficiencies (Hasegawa et al. 2000; Munns

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and Tester 2008; Yeo 1998). There were numerous reports about salt stress to growth and development of sorghum. Germination rate and seedling vigor of sorghum was significantly reduced under salt stress conditions (Almodares et al. 2007; Rani et al. 2012). Salt strongly inhibited plant growth and decreased stem dry weight with the increase of salt stress (Netondo et al. 2004a, b). Salinity reduced relative growth rates and increased soluble carbohydrates (Lacerda et al. 2005). Total soluble sugar increased in stalk of sorghum with increasing salinity level (Almodares et al. 2008). There were significant differences between genotypes of sorghum grown in salt stress conditions (Maiti et al. 1994).

With advent of molecular markers technology, many studies had been conducted to identify genes or quantitative trait loci (QTLs) affecting salt tolerance in different plant species during different developmental stages. Identification of markers tightly linked to genes or QTLs for salt tolerance would help to develop salt-tolerant varieties by marker assisted selection (MAS), and increase breeding efficiency (Lin et al. 2004; Lee et al. 2003). QTLs for salt tolerance had been detected in gramineous crops such as rice (Prasad et al. 2000; Lin et al. 2004; Lee et al. 2006; Wang et al. 2012), wheat (Ma et al. 2007; Xu et al. 2012) and barley (Mano and Takeda 1997; Zhou et al. 2012). One major QTL for shoot K^+ concentration in rice had been cloned and defined *SKCI*, which was involved in regulating K^+/Na^+ homeostasis under salt stress (Ren et al. 2005). Another major *RASI* QTL for percentage of green seedlings cloned in *Arabidopsis* (Ren et al. 2010), was an ABA and salt stress-inducible gene encoding a previously undescribed plant-specific protein. Above information will help to understand the genetic mechanism for salt tolerance of other crops and engineer higher salt tolerant crop by MAS.

Molecular markers associated with salt stress of sorghum had been detected in productive growth stage following bulked segregant analysis (Younis et al. 2007). Genetic variability among seven sorghum cultivars under salt stress had been investigated, and molecular markers for salt tolerance in some sorghum genotypes had been detected (Khalil 2013). However, to our knowledge there had been no report on QTL analysis for salt tolerance at germination and seedling stage in sorghum. Seed germination plays an important role in uniform seedling emergence and vigorous

stand establishment. It is more salt tolerant at germination stage than at later stages of growth (Francois et al. 1984). Macharia et al. (1994) reported salinity caused more serious damage at seedling stage than in any other stage in sorghum. To get a better understanding of genetic control and to improve salt tolerance of sorghum, 181 recombinant inbred lines (RILs) were evaluated for salt tolerance at germination and seedling stage, respectively. QTLs for traits related to salt tolerance at both stages were analyzed. The results would help to understand genetic control for salt tolerance at seed germination and seedling stage, and improve salt tolerance of sorghum by MAS.

Materials and methods

Plant materials

One hundred and eighty-one RILs (F_7) derived from cross of grain sorghum (Shihong137) and sweet sorghum (L-Tian) were used for this study. Each RIL was derived from a single F_2 plant following single seed descent (SSD) until F_7 . Shihong137 is a dwarf grain sorghum inbred line of China and salt-tolerant. L-Tian is a sweet sorghum inbred line of China and sensitive to salt stress. The traits were uniform in family, and there were larger variations between families.

Evaluation for salt tolerance

Procedure of evaluation for salt tolerance was performed according to Lu (2006) with minor modifications. In November 2011, preliminary experiment of two parents and ten lines randomly selected from 181 RILs were conducted for salt tolerance with five NaCl concentrations of 0.0, 1.0, 1.5, 2.0 and 2.5 % at germination stage. Compared to control, a large variance of germination vigor and germination percentage was found between parents and ten lines at 2.0 % NaCl solution. Finally, the concentration of 2.0 % NaCl was determined for the screen of salt tolerance. The experiment was repeated twice with the same method.

Fifty grains of each parent and RILs were surface-sterilized with 1.0 % sodium hypochlorite solution for 10 min, and then were rinsed three times with sterile distilled water. The seeds were placed on petri dishes

(Diameter = 12 cm) with two layers of filter paper moistened with 6 mL distilled water in control and 6 mL 2.0 % NaCl solution in treatment. Each petri dish was sealed with sealing membrane to prevent evaporation. The germination test was performed under a light/dark photoperiod of 12 h/12 h with a day/night temperature of 29/22 °C in growth chambers with three replications. Germination was defined as at least 1 mm of radicle or plantule emerged. 3 days after incubation, germinated seeds were counted, and germination vigor was recorded. Then distilled water was added to maintain NaCl concentration and control volume, respectively. Petri dishes were sealed again, and continuously incubated in growth chambers. Germination percentage was recorded until the seventh day. Relative salt-injury rate (RSR) was applied to evaluate parents and RILs for salt tolerance. Low RSR represents high salt resistance, and vice versa. It was calculated by following formula: $RSR (\%) = 100 \times \sum (G_c - G_i) / 3G_c$, where G_c represents germination percentage for control (%), and G_i represents germination percentage for treatment (%).

0.6 % NaCl solution was determined for salt tolerance evaluation at seeding stage based on preliminary experiment (Wang et al. 2013). In May 2012, 0.0 and 0.6 % NaCl solution was applied for 181 RILs and two parents in randomized complete blocks with three replications. 100 seeds of each line were placed on petri dish with distilled water for germination at room temperature. Eight uniformly germinated seeds per line of each replication were transplanted on thin styrofoam board with a nylon net bottom in a plastic box with distilled water. Seedlings floated on water to two leaves stage, then distilled water replaced with Hoagland solution containing 0.0 and 0.6 % NaCl. The solution was refreshed every 5 days, and NaCl concentration was maintained at 0.6 % every day by adding certain volume of distilled water. After 15 days of salt stress, based on injury degree of seedlings and number of green leaves, salt tolerance level was divided into 0, 1, 2, 3, 4 and 5. Salt injury index was used to evaluate salt tolerance based on following formula: Salt injury index (%) = $100 \% \times [\sum KiNi / (5 \sum Ni)]$, where K_i represents salt tolerance level, and N_i represents seedling number of each salt tolerance level.

Three uniform seedlings per line were selected to harvest separately after evaluation for salt tolerance. Length and fresh weight of shoot and root were investigated. Then shoot and root were oven-dried to

obtain dry weight. The experiment was repeated twice with the same method.

SSR markers and genetic map construction

A total of 616 SSR markers including 141 polymorphic markers between parents in our previous study (Guan et al. 2011) and 475 new SSR markers with known chromosome positions (Satish et al. 2009; Srinivas et al. 2009; Yonemaru et al. 2009) were collected in our study. In July 2011, seeds harvested from plants of 2010 were planted in petri dishes at room temperature, and seedlings at 4–5 leaves stage were collected for DNA extraction and SSR analysis. Total genomic DNA was extracted according to the method described by Dellaporta et al. (1983). PCR amplification of SSR loci was performed in a 10 µL reaction mixture containing 25 ng/µL template DNA 1 µL, 5 µM forward primer 1 µL, 5 µM reverse primer 1 µL, ddH₂O 2 µL and 2 × Taq Plus PCR MasterMix 5 µL (Tiangen Biotec Co, Beijing, China) in a Thermal Cycler (Biometa, Germany). The annealing temperature was 55 °C for most primers. PCR products were separated in a DNA sequencing electrophoresis apparatus in 6.0 % polyacrylamide gels containing 1 × TBE buffer at 60 W of constant power. The DNA fragments were visualized by silver staining and scored either parental (1 or 2), heterozygous (3), or missing data for (0). The detail procedure for SSR analysis was described in our previous study (Guan et al. 2011).

Linkage groups (LGs) were constructed using Mapmaker/EXP3.0 (Lander et al. 1987; Lincoln et al. 1993). Informativeness criteria of 4.0 and 181 was used to define a highly-informative marker. Linkage criteria of 3.0 and 50.00 was used in the “assign” command. MapChart 2.2 was used to graphically draw LGs of genetic map (Voorrips 2002).

QTL mapping

Statistical analysis of phenotypes was performed using SPSS 16.0 software (SPSS Inc, Chicago, USA). QTLs for each trait were analyzed using composite interval mapping (CIM) with the software of Cartographer 2.5 (Wang et al. 2010). The threshold for significant QTLs was determined by 1,000 permutation test at 0.05 probability level (Churchill and Doerge 1994). The location of a QTL was described according to its LOD

Table 1 Phenotypic performance for traits related to salt tolerance among parents and RILs under control and salt stress

Trait	Treatment	Parents ^a		RILs ^b				
		L-Tian	Shihong137	Mean	Max.	Min.	SD	CV
RSR (%)		51.9 ± 1.21	19.3 ± 1.25**	33.11	91.7	0.7	20.1	0.61
GV (%)	Control	90.0 ± 1.00	100.0 ± 0.00**	87.11	100.00	42.0	9.8	0.11
	2.0 %NaCl	0.0 ± 0.00	70.0 ± 1.30**	14.15	85.30	0.0	17.3	1.22
GP (%)	Control	90.0 ± 1.00	100.0 ± 0.00**	89.18	100.00	42.0	9.3	0.10
	2.0 %NaCl	43.3 ± 1.78	80.7 ± 1.78**	60.3	94.00	6.7	20.5	0.34
SII (%)		78.3 ± 3.21	38.3 ± 1.98**	63.8	88.3	13.3	14.2	0.22
SH (cm)	Control	33.33 ± 0.58	38.67 ± 4.86	35.76	49.57	13.07	7.08	0.20
	0.6 %NaCl	10.08 ± 0.47	18.11 ± 2.09*	13.79	21.93	8.37	2.76	0.20
RL (cm)	Control	10.00 ± 1.00	14.20 ± 3.46	12.72	17.17	6.83	1.61	0.13
	0.6 %NaCl	9.86 ± 1.22	13.87 ± 1.67*	11.02	15.63	7.43	1.26	0.11
SFW (g)	Control	4.05 ± 0.18	6.66 ± 0.25**	4.68	12.86	0.84	1.96	0.42
	0.6 %NaCl	0.52 ± 0.01	2.16 ± 0.65*	1.19	2.93	0.29	0.57	0.47
RFW (g)	Control	1.69 ± 0.16	2.88 ± 0.17**	2.35	6.06	0.66	0.10	0.43
	0.6 %NaCl	0.60 ± 0.08	1.57 ± 0.33*	1.03	2.47	0.33	0.38	0.37
TFW (g)	CK	5.74 ± 0.17	9.54 ± 0.29**	7.03	18.21	1.66	2.86	0.41
	0.6 %NaCl	1.13 ± 0.07	3.73 ± 0.96*	2.24	5.12	0.68	0.89	0.40
SDW (g)	Control	0.31 ± 0.05	0.52 ± 0.04**	0.37	0.84	0.08	0.14	0.39
	0.6 %NaCl	0.07 ± 0.01	0.21 ± 0.06*	0.12	0.28	0.03	0.05	0.44
RDW (g)	Control	0.10 ± 0.02	0.19 ± 0.02**	0.14	0.88	0.05	0.09	0.64
	0.6 %NaCl	0.05 ± 0.01	0.13 ± 0.04*	0.07	0.15	0.01	0.03	0.35
TDW (g)	Control	0.40 ± 0.05	0.69 ± 0.07**	0.51	1.21	0.13	0.20	0.38
	0.6 %NaCl	0.12 ± 0.01	0.33 ± 0.09*	0.19	0.40	0.07	0.08	0.39

RSR relative salt-injury rate, GV germination vigor, GP germination percentage, SII Salt injury index, SH shoot height, RL root length, SFW shoot fresh weight, RFW root fresh weight, TFW total fresh weight, SDW shoot dry weight, RDW root dry weight, TDW total dry weight, SD standard deviation, CV coefficient of variation

*, ** Significant at 0.05 and 0.01 level, respectively

^a Mean ± SD (standard deviation)

^b RILs sample size $n = 181$, replications $r = 3$

peak and the flanking region with 95 % confidence interval.

Results

Statistical analysis of phenotypes

Statistical analysis of phenotypes for traits related to germination and seedling growth under control and salt stress of RILs and their parents were summarized in Table 1. There were significant differences for all traits between two parents under control and NaCl stress, except shoot height and root length under control. Shihong137 was superior to L-Tian for all traits related

to germination and seedling growth. There was a continuous frequency distribution and transgressive segregation for all traits among RILs under control and salt treatment. The coefficient of variation was higher than 10 % for all traits. Salt stress effects were observed for all traits, and mean values of RILs under control were higher than that of in salt stress. Correlation analysis showed that there were positive correlations between traits at seedling stage under control and 0.6 % NaCl solution (Table 2). Correlation coefficients were all significant, and in very good accordance under two treatments. Correlation coefficients were higher between shoot height and total fresh weight and total dry weight than that of root length with total fresh weight and total dry weight in both conditions, respectively.

Table 2 Correlation coefficients of different traits related to salt tolerance under control (below diagonal) and 0.6 % NaCl (above diagonal) stress at seedling stage

Trait	SH	RL	SFW	RFW	SDW	RDW	TFW	TDW
SH		0.31**	0.91**	0.73**	0.91**	0.66**	0.85**	0.89**
RL	0.21**		0.34**	0.45**	0.30**	0.36**	0.38**	0.34**
SFW	0.89**	0.21**		0.80**	0.94**	0.73**	0.93**	0.93**
RFW	0.75**	0.29**	0.85**		0.77**	0.74**	0.91**	0.82**
SDW	0.78**	0.19**	0.90**	0.78**		0.72**	0.87**	0.97**
RDW	0.30**	0.12	0.38**	0.40**	0.39**		0.75**	0.86**
TFW	0.87**	0.25**	0.98**	0.93**	0.89**	0.40**		0.89**
TDW	0.71**	0.20**	0.83**	0.75**	0.91**	0.74**	0.83**	

SH shoot height, RL root length, SFW shoot fresh weight, RFW root fresh weight, TFW total fresh weight, SDW shoot dry weight, RDW root dry weight, TDW total dry weight

*, ** Significant at 0.05 and 0.01 level, respectively

Construction of genetic linkage map

Two hundred and forty-seven SSR markers showed polymorphism between two parents. 66 markers were excluded due to ambiguous bands in genotyping, serious deviation from normal segregation or not linked to any group in linkage analysis. The Chi square test for 181 polymorphic markers indicated that 140 co-dominant markers segregated in Mendelian fashion. 41 markers (22.7 %) deviated from normal segregation with Chi square values over 5.99. The final genetic map was built using 181 SSR markers, which consisted of ten LGs, covering 2415.3 cM with an average distance of 13.34 cM between markers. SSR locations were compared to previously published sorghum maps (Bhattaramakki et al. 2000; Menz et al. 2002; Wu and Huang 2006; Li et al. 2009; Mace et al. 2009; Yonemaru et al. 2009) for assigning LGs to sorghum chromosomes SBI-01 to SBI-10 (Kim et al. 2005) (Fig. 1).

QTLs analysis at germination stage

Three QTLs for germination vigor were detected on SBI-02 and SBI-03 under control. The phenotypic variation explained (PVE) ranged from 5.4 to 6.0 %, and additive effect could increase germination vigor by 2.33–2.55 %. Three QTLs on SBI-01 and SBI-04 were identified for germination vigor under 2.0 % NaCl with PVE of 5.3–6.5 %, and additive effect of 4.09–5.03 % (Table 3; Fig. 1).

Three QTLs for germination percentage under control were detected on SBI-01, SBI-02 and SBI-

07, and could explain 5.2–8.9 % of phenotypic variation. *qGP2* was with the nearest genetic distance to Xcup26, and the positive effect was from L-Tian, which increased germination percentage by 2.3 %. The positive effect of other two QTLs was from the parent of Shihong137. Two QTLs associated with germination percentage in 2.0 % NaCl were identified on SBI-07 with PVE of 10.0 and 9.0 %, respectively. The positive alleles from Shihong137 and L-Tian could enhance germination percentage by 6.55 and 6.58 %, respectively.

Only one QTL for RSR was found on SBI-07 with the PVE of 8.7 %. The negative effect from L-Tian could decrease RSR by 6.34 %.

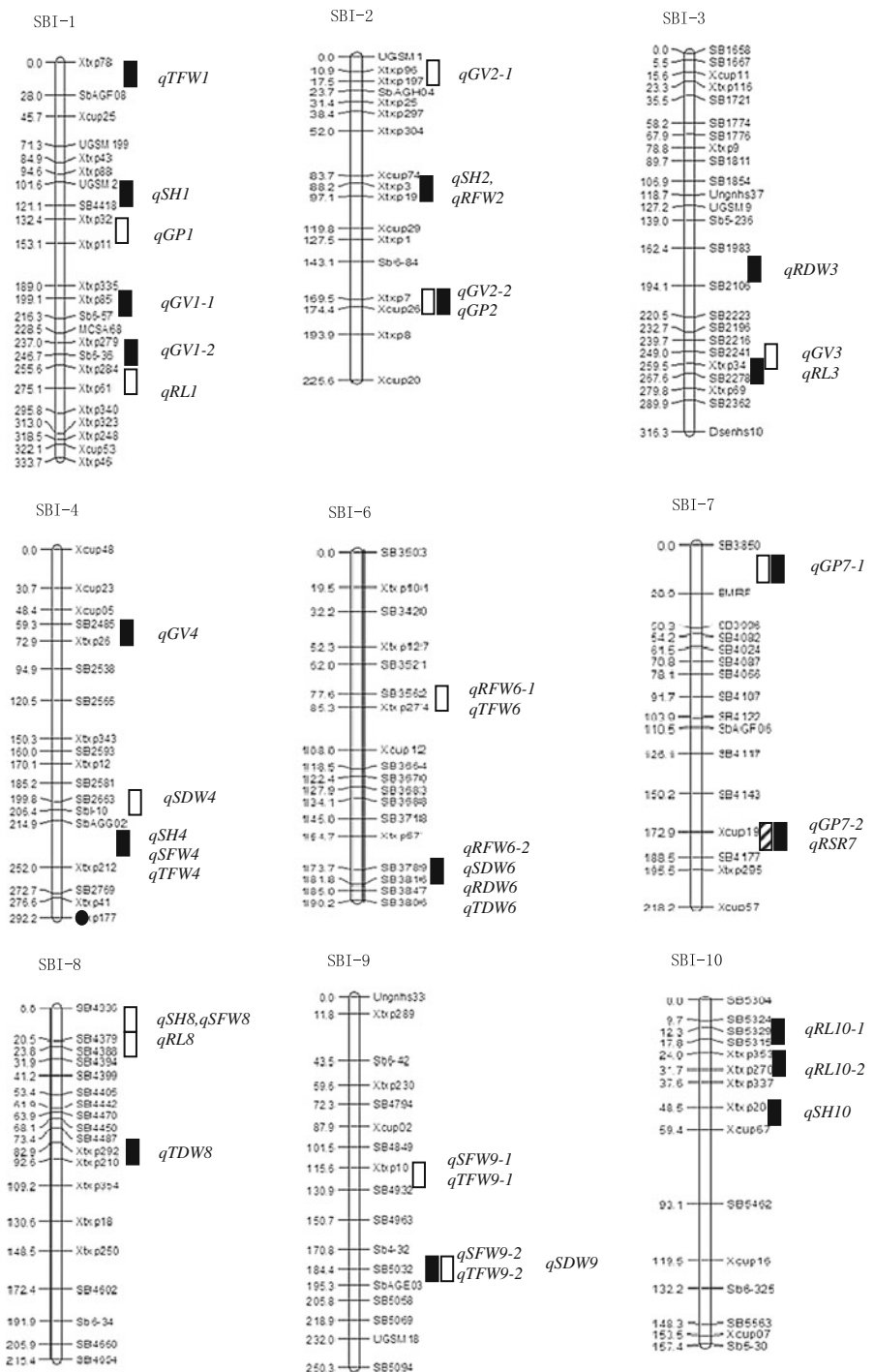
QTLs analysis at seedling stage

A total of 29 QTLs for nine traits at seedling stage were detected on eight chromosomes under two treatments. The QTLs explained 5.3–21.9 % of the phenotypic variation. None QTL was found for salt injury index, root dry weight and total dry weight under control (Table 3; Fig. 1).

One QTL in control and four QTLs in salt stress were detected for shoot height on SBI-01, SBI-02, SBI-04 SBI-08 and SBI-10 accounting for 6.1–15.6 % of phenotypic variation. The positive effects were all from Shihong137 except that of *qSH10*, and could increase shoot height 0.70–1.83 cm.

There were three QTLs in control and two QTLs in salt stress for root length identified on SBI-01, SBI-03, SBI-08 and SBI-10 with PVE of 5.3–16.0 % and additive effect of 0.29–0.75 cm.

Fig. 1 Locations of QTLs for traits related to salt tolerance in control and salt stress at germination and seedling stages based on RILs derived from Shihong137 $\times$ L-Tian



Four QTLs on SBI-04, SBI-08 and SBI-09 for shoot fresh weight were detected, and account for 5.5–11.65 % of the phenotypic variation. Three QTLs for root fresh weight were located on SBI-02 and SBI-06 with PVE ranged from 5.38 to 6.87 %.

Five QTLs controlling total fresh weight were identified with PVE of 5.6–21.9 %. *qTFW1* on marker interval of Xtp78-SbAGF08 could explain 21.9 % of phenotypic variation and the allele from Shihong137, which could increase total fresh weight by 0.49 g.

Table 3 QTLs detected for traits related to salt tolerance under control and salt stress at both stages

Trait	Treatment	QTL	Site (cM) ^a	Marker interval ^b	LOD	Additive ^c	PVE (%) ^d
RSR		<i>qRSIR7</i>	7.60	Xcup19- SB4177	2.45	6.34	8.7
GV	Control	<i>qGV2-1</i>	0.00	Xtxp96 -Xtxp197	2.65	2.33	5.4
		<i>qGV2-2</i>	0.83	Xtxp7- Xcup26	2.65	−2.55	5.9
		<i>qGV3</i>	0.46	SB2241- Xtxp34	2.89	−2.43	6.0
		<i>qGV1-1</i>	0.03	Xtxp85-Sb6_57	2.50	4.09	5.3
	2.0 % NaCl	<i>qGV1-2</i>	0.87	Sb6_36- Xtxp284	2.85	−4.52	6.5
		<i>qGV4</i>	4.05	SB2485 -Xtxp26	2.50	−5.03	6.4
GP	Control	<i>qGP1</i>	6.68	Xtxp32 -Xtxp11	2.63	2.53	7.3
		<i>qGP2</i>	0.03	Xtxp7- Xcup26	2.58	−2.30	5.2
		<i>qGP7-1</i>	6.01	SB3850 -BMR	3.06	2.78	8.9
	2.0 % NaCl	<i>qGP7-1</i>	8.01	SB3850 -BMR	2.74	6.55	10.0
		<i>qGP7-2</i>	5.59	Xcup19- SB4177	2.75	−6.58	9.0
		None					
SII		None					
SH	Control	<i>qSH8</i>	0.01	SB4336 -SB4379	3.09	1.83	6.4
	0.6 % NaCl	<i>qSH1</i>	7.51	UGSM2- SB4418	4.30	1.04	13.5
		<i>qSH2</i>	0.07	Xtxp3 -Xtxp19	3.07	0.70	6.1
		<i>qSH4</i>	1.98	SbAGG02 -Xtxp212	2.91	1.12	15.6
		<i>qSH10</i>	0.86	Xtxp20- Xcup67	3.28	−0.75	7.0
		<i>qRL1</i>	0.03	Xtxp284 -Xtxp61	3.18	0.48	8.9
RL	Control	<i>qRL8</i>	0.07	SB4379 -SB4388	2.72	−0.39	5.5
		<i>qRL3</i>	0.04	Xtxp34 -SB2278	2.62	−0.29	5.3
	0.6 % NaCl	<i>qRL10-1</i>	0.01	SB5329- SB5315	2.89	−0.33	6.7
		<i>qRL10-2</i>	1.66	Xtxp353- Xtxp270	5.86	−0.75	16.0
		<i>qSFW8</i>	0.01	SB4336 -SB4379	2.62	0.48	5.5
		<i>qSFW9-1</i>	0.07	Xtxp10 -SB4932	3.31	−0.52	7.0
SFW	Control	<i>qSFW8</i>	0.01	SB4336 -SB4379	2.62	0.48	5.5
	0.6 % NaCl	<i>qSFW4</i>	9.98	SbAGG02 -Xtxp212	2.90	0.19	11.6
		<i>qSFW9-2</i>	2.02	SB5032 -SbAGE03	3.17	−0.14	7.1
RFW	Control	<i>qRFW6-1</i>	0.01	SB3562- Xtxp274	2.55	0.24	5.4
	0.6 % NaCl	<i>qRFW2</i>	0.07	Xtxp3 -Xtxp19	3.40	0.10	6.9
		<i>qRFW6-2</i>	0.02	SB3789- SB3816	3.09	0.10	6.3
TFW	Control	<i>qTFW6</i>	0.01	SB3562- Xtxp274	2.68	0.70	5.6
		<i>qTFW9-1</i>	0.07	Xtxp10 -SB4932	2.73	−0.69	5.7
	0.6 % NaCl	<i>qTFW1</i>	14.01	Xtxp78 -SbAGF08	3.55	0.49	21.9
		<i>qTFW4</i>	11.98	SbAGG02 -Xtxp212	2.63	0.31	11.5
		<i>qTFW9-2</i>	2.02	SB5032 -SbAGE03	3.19	−0.25	7.4
SDW	Control	<i>qSDW4</i>	0.02	Sbl_10- SbAGG02	3.08	0.04	6.3
		<i>qSDW9</i>	4.02	SB5032 -SbAGE03	2.54	−0.04	6.4
	0.6 % NaCl	<i>qSDW6</i>	4.02	SB3789- SB3816	2.51	0.01	6.0
RDW	Control	None					
	0.6 % NaCl	<i>qRDW3</i>	0.02	SB1983 -SB2106	2.60	0.01	5.4
		<i>qRDW6</i>	2.02	SB3789- SB3816	2.56	0.01	6.1
TDW	Control	None					
	0.6 % NaCl	<i>qTDW6</i>	4.02	SB3789- SB3816	3.40	0.02	8.1

Table 3 continued

Trait	Treatment	QTL	Site (cM) ^a	Marker interval ^b	LOD	Additive ^c	PVE (%) ^d
		<i>qTDW8</i>	0.06	Xtxp292 -Xtxp210	2.68	0.02	5.5

RSR relative salt-injury rate, *GV* germination vigor, *GP* germination percentage, *SII* Salt injury index, *SH* shoot height, *RL* root length; *SFW* shoot fresh weight, *RFW* root fresh weight, *TFW* total fresh weight, *SDW* shoot dry weight, *RDW* root dry weight, *TDW* total dry weight

^a Means the nearest distance of *F* value peak for QTL to the marker in the marker interval

^b Bold font markers are those closer to the putative QTL

^c Positive values indicate that alleles from Shihong137, negative values indicate that alleles from L-Tian

^d Phenotypic variance explained by QTL

Three QTLs for shoot dry weight were detected with PVE of 6.3, 6.4 and 6.0 %, respectively. Two QTLs controlling shoot dry weight and total dry weight were detected in 2.0 % NaCl. The PVE was 5.4 and 8.1 %, and positive effects were all from salt tolerant parent Shihong137.

Discussion

Sorghum (*S. bicolor* (L.) Moench] is often grown on saline and alkaline land as well as barren land in China. Shandong is one of the provinces with large saline areas in China. There are mainly three types of coastal saline soils: slight saline soils (Salt content = 0.2–0.4 %), middle saline soils (Salt content = 0.4–0.7 %) and serious saline soils (Salt content = 1.0 %) (Zhu and He 1985). Sorghum is moderately salt tolerant and could grow at 0.3–0.6 % NaCl concentrations. There were various genotypes existing among sorghum in response to salinity stress (Netondo et al. 2004a, b). Therefore, breeding and cultivation of high-yielding salt-tolerant sorghum varieties is one of potential strategies to use saline soils. Sorghum is more salt tolerant at germination than at seedling and adult stages (Francois et al. 1984). As reported by Francois et al. (1984), significant difference of germination percentage between varieties was detected at higher salt concentration. The similar result was obtained in our study. The significant variation between parents and 181 lines were identified in 2.0 % NaCl solution at germination stage. However, Seedling stage of sorghum was the most sensitive stage to salt injury (Macharia et al. 1994). In our research, seedling could survive and also showed different degree of salt injury at 0.6 % NaCl. Furthermore, this concentration is approximately equal to the

salinity concentration in the soil/field conditions where sorghum is grown in China. With the growth of plants, salt resistance was increased (Foolad 2004; Cuartero et al. 2006). Salt tolerance at seedling stage was consistent with the adult stage (Azhar and McNeilly 1987). Therefore, genetic information at seedling stage would help to develop sorghum varieties of salt tolerance.

Salinity could modify physiological and biochemical processes of plant (Dubey 1994). The ability to withstand salt stress is a developmentally regulated, stage-specific and environment-specific phenomenon (Ashraf and Foolad 2013). Genetic analysis for salt tolerance at single developmental stage might simplify the underlying genetic components. Previous studies demonstrated that salinity caused reduction in germination (Igartua et al. 1994) and growth of sorghum (Maiti et al. 1994). Compared to results in control, seed germination was significantly inhibited, and germination percentage was reduced under salt stress in our study. At seedling stage, growth of all RILs and parents was inhibited, leaves were dehydrated and wilted to some extent, and roots were damaged and became brown gradually since salt stress beginning. Above results indicated that physiological mechanisms of sorghum was modified under salt stress.

Accurate establishment and evaluation of phenotype for salt tolerance is the most crucial step in QTL mapping. Characters such as average shoot length (Mano and Takeda 1997), salt tolerance index, fresh and dry weight of radical and plantule (Ma et al. 2007) and seed imbibition rate and germination percentage (Wang et al. 2011) at germination stage, and shoot height, root length, dry weights of shoots and roots at seedling stage (Wang et al. 2011; Xu et al. 2012) had been used to evaluate for salt tolerance, and conducted for QTL analysis. In the present study, 181 RILs and

two parents were investigated for salt tolerance at germination and seedling stage with 12 traits most mentioned above. At germination stage, One QTL for relative salt-injury rate on SBI-7 was identified. Three QTLs of germination vigor were detected under control and 2.0 % NaCl, respectively. Three and two QTLs for germination percentage were detected under control and 2.0 % NaCl, respectively. Among them, only *qGP7-1* was found simultaneously under control and 2.0 % NaCl. More QTLs were found under salt stress than control at seedling stage. 19 QTLs were mapped, and 6 QTLs were mapped with PVE more than 10.0 % under salt stress (Table 3; Fig. 1). It was found that almost all QTLs for traits related to salt tolerance were adaptive and salt-specific expressed. In general, number and PVE of QTLs detected under salt stress were larger than that of QTLs identified under control. We predicted that genetic mechanism for salt tolerance at germination and seedling stage might be different.

QTLs analysis for salt tolerance had been reported in several plants, especially in cereal model plant of rice (Prasad et al. 2000; Lin et al. 2004; Lee et al. 2006; Wang et al. 2012). There were several major QTLs mapped with large PVE, and some had been cloned. *qSKC-1* for shoot K^+ concentration, explaining 40.1 % of total phenotypic variance, were mapped and cloned in rice (Ren et al. 2005). For percentage of green seedlings of *Arabidopsis*, one major QTL with PVE of 76.6 % was cloned (Ren et al. 2010). In our study, though none major QTL was detected at germination stage, six major QTLs with PVE more than 10.0 % were detected at seedling stage under salt stress. Two major QTLs were found for shoot height with PVE of 13.5 and 15.6 % and positive effect from Shihong137, which could increase shoot height 1.04 and 1.12 cm. *qRL10-2* was a major QTL controlling root length with additive effect from L-Tian. *qSFW4* and *qTFW4* were two major QTLs controlling shoot fresh weight and total fresh weight, respectively and located on a same marker interval. One major QTL for total fresh weight was mapped on SBI-01, and account for 21.9 % of the phenotypic variation. These major QTLs could be further applied in improving salt tolerance by MAS, and isolating genes for salt tolerance at seedling stage of sorghum.

Co-localized QTLs controlling more than one trait were identified in wheat, barley and rice in salt tolerance study (Ma et al. 2007; Mano and Takeda

1997; Wang et al. 2011). The phenomena of QTLs co-localization were reported from previous studies in sorghum as well. *Dw2* controlling plant height of sorghum on SBI-06 was co-localized with a photoperiodic sensitive locus (Lin et al. 1995). QTLs controlling plant height of sorghum were localized at the same chromosome regions with QTLs for shoot and leaf fresh weight and juice weight in our previous study using F_2 and $F_{2:3}$ populations derived from the same cross (Guan et al. 2011). Five chromosome regions were found controlling more than one trait simultaneously in present study. *qGP7-2* and *qRSR7* were located on a same marker interval of Xcup19-SB4177. The additive effect from the same parent of L-Tian could decrease relatively salt-injury rate and increase germination percentage, respectively. *qSH2* and *qRFW2* were co-located on SBI-02, and the genetic distance to the nearest marker Xtxp3 was 0.07 cM. QTLs for shoot fresh weight and total fresh weight were identified and mapped on a same marker interval of SB5032-SbAGE03. Chromosome region controlling three traits was found on SBI-04. *qRFW6*, *qSDW6*, *qRDW6* and *qTDW6* were mapped on the marker interval of SB3789-SB3816 with additive effect from a same parent Shihong137. These QTLs might be single gene with pleiotropic effect or tightly linked gene with different function. Co-localization of QTLs with uniform additive effect source made it possible for coordinated improvement and increasing efficiency of MAS. Therefore, the five co-localized chromosome regions could help to develop varieties with high salt tolerance in sorghum production.

Acknowledgments This work was supported by National Science and Technology Pillar Program from Ministry of Science and Technology of China (2009BADA7B01), Scientific Research Foundation for Outstanding Young Scientists of Shandong Province (BS2011NY019) and the Earmarked Fund for China Agriculture Research System (CARS-06).

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